

A nonzero fixed-product preserver that destroys invertibility

Candidate counterexample for arXiv:2201.12912, Problem 3.5

11 August 2026

Status. Candidate full counterexample, likely valid, requiring human review. The example is a separable commutative unital complex Banach algebra; the map is a bounded linear involution; and the prescribed elements satisfy $c = d \neq 0$ and $c^2 = c$.

1 Source question

Julius asks the following in Problem 3.5 of [1], on PDF page 6:

If ϕ satisfies the hypotheses of Theorem 3.3, does $d \neq 0$ imply that ϕ preserves invertibility?

Here the hypotheses are that $\mathcal{A}$ is a unital Banach algebra with dense invertible group and that $\phi : \mathcal{A} \rightarrow \mathcal{A}$ is a bijective continuous linear map satisfying

$$ab = c \implies \phi(a)\phi(b) = d$$

for fixed $c, d \in \mathcal{A}$.

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Consider the bijective map $f(a) = a^{-1}c$ on the group of invertible elements. Hence $f(B_\varepsilon(s))$ is another open set consisting entirely of invertible elements of the form $a^{-1}c$, where $\|a - s\| < \varepsilon$. Another application of Lemma 3.2 gives the existence of an element $t \in B_\varepsilon(s)$ such that $\phi(t^{-1}c)$ is also invertible. Now $\phi(t)$ and $\phi(t^{-1}c)$ are both invertible and their product is d , thus d is invertible. Injectivity is not required. $\square$

Unless X is finite-dimensional, $\mathcal{B}(X)$ does not generally have a dense invertible group. Regrettably Theorem 3.3 therefore does not apply to perhaps the most interesting cases. Nevertheless, it seems unlikely that the conclusions of the theorem should fail.

Problem 3.4. Does Theorem 3.3 hold if $\mathcal{A} = \mathcal{B}(X)$, where X is an infinite-dimensional Banach space?

In addition, Costara shows something stronger with [Lemma 1, [15]] for the Banach algebra of $n \times n$ matrices $M_n(\mathbb{C})$; in particular, that $d \neq 0$ implies that ϕ preserves invertibility. The proof assumes that there is an invertible element t_0 such that $\phi(t_0)$ is not invertible. The argument works just as well for arbitrary Banach algebras with dense invertible group until the ultimate step when Costara uses convenient matrix representations of noninvertible elements to contradict the assumption on $\phi(t_0)$. We suspect that a similar result will hold for Theorem 3.3.

Problem 3.5. If ϕ satisfies the hypotheses of Theorem 3.3, does $d \neq 0$ imply that ϕ preserves invertibility?

Figure 1: The context and exact statement of Problem 3.5 in [1, p. 6]. This image was rendered and cropped from the local source paper PDF.

2 Proof intuition

The condition becomes very weak at a zero product in an integral domain: $ab = 0$ then forces one factor to be zero, so any linear map automatically takes that pair to another zero product. A direct product supplies a separate coordinate in which the prescribed product is nonzero. Thus the two desired features can coexist: the first coordinate guarantees $d \neq 0$, while the domain coordinate leaves enough linear freedom to send a unit to a nonunit.

What remains is to produce a unital Banach integral domain with dense units. We do this explicitly. A rapidly decreasing convolution weight makes a nonunital weighted power-series domain radical. In its unitization, an element is invertible exactly when its scalar coordinate is nonzero, so the units are dense. Swapping the unit with the first power-series monomial is then a bounded linear involution that does not preserve invertibility.

3 A dense-unit Banach integral domain

Put $\rho_n = e^{-n^2}$ for $n \geq 1$, and let

$$\mathcal{R} = \left\{ a = (a_n)_{n \geq 1} : \|a\|_\rho := \sum_{n \geq 1} |a_n| \rho_n < \infty \right\}.$$

Give $\mathcal{R}$ the Cauchy convolution product

$$(a * b)_n = \sum_{j=1}^{n-1} a_j b_{n-j}.$$

Let $z = (1, 0, 0, \dots)$, so z^n is the unit vector in coordinate n .

Lemma 1. *$\mathcal{R}$ is a separable commutative radical Banach algebra and an integral domain.*

Proof. Since

$$\rho_{m+n} = e^{-(m+n)^2} \leq e^{-m^2} e^{-n^2} = \rho_m \rho_n,$$

Tonelli's theorem gives

$$\|a * b\|_\rho \leq \sum_{m,n \geq 1} |a_m| |b_n| \rho_m \rho_n = \|a\|_\rho \|b\|_\rho.$$

Completeness and separability follow from weighted ℓ_1 .

If $a, b \neq 0$, let m and n be their least nonzero indices. The coefficient of $a * b$ at $m + n$ is exactly $a_m b_n \neq 0$. Hence $\mathcal{R}$ has no zero divisors.

It remains to check radicality. The spectral-radius formula gives

$$r(z) = \lim_{k \rightarrow \infty} \|z^k\|_\rho^{1/k} = \lim_{k \rightarrow \infty} e^{-k} = 0.$$

By spectral mapping, every polynomial $p(z)$ with $p(0) = 0$ has spectral radius zero. Such polynomials (the finitely supported sequences) are dense in $\mathcal{R}$. Since $\mathcal{R}$ is commutative, the spectral radius is subadditive; thus, for any $a \in \mathcal{R}$ and any such polynomial p ,

$$r(a) \leq r(a - p) + r(p) \leq \|a - p\|_\rho.$$

Letting $p \rightarrow a$ yields $r(a) = 0$. Every element is therefore quasinilpotent, which says precisely that the commutative Banach algebra $\mathcal{R}$ is radical. Standard spectral-radius facts used here may be found in [2]. $\square$

Let $\mathcal{A} = \mathcal{R}^\# = \mathbb{C}1 \oplus \mathcal{R}$ be the unitization, with norm $\|\lambda 1 + r\| = |\lambda| + \|r\|_\rho$.

Lemma 2. $\mathcal{A}$ is a separable commutative unital Banach integral domain, and

$$\mathcal{A}^{-1} = \{\lambda 1 + r : \lambda \neq 0\}.$$

In particular, its invertible group is dense.

Proof. If $\lambda \neq 0$, radicality of $\mathcal{R}$ implies that $1 + \lambda^{-1}r$ is invertible in the unitization, so $\lambda 1 + r$ is invertible. If $\lambda = 0$, the quotient homomorphism $\mathcal{A} \rightarrow \mathcal{A}/\mathcal{R} \cong \mathbb{C}$ sends r to zero, so r cannot be invertible. This proves the displayed characterization and density follows by perturbing the scalar coordinate.

For domainhood, suppose $xy = 0$. If either factor has nonzero scalar coordinate, it is a unit and the other factor is zero. If both scalar coordinates vanish, then $x, y \in \mathcal{R}$, and Lemma 1 gives $x = 0$ or $y = 0$. $\square$

Let

$$\mathcal{R}_0 = \{(a_n) \in \mathcal{R} : a_1 = 0\}.$$

The first-coordinate functional is bounded, so $\mathcal{A} = \mathbb{C}1 \oplus \mathbb{C}z \oplus \mathcal{R}_0$ is a topological direct sum. Define

$$L(\lambda 1 + \mu z + r_0) = \mu 1 + \lambda z + r_0. \quad (1)$$

Then L is complex-linear, $L^2 = I$, and it is bounded: indeed,

$$\|L(\lambda 1 + \mu z + r_0)\| = |\mu| + e^{-1}|\lambda| + \|r_0\|_\rho \leq e(|\lambda| + e^{-1}|\mu| + \|r_0\|_\rho).$$

In particular, L is a Banach-space automorphism. But $L(1) = z$ is not invertible by Lemma 2.

4 The counterexample

Theorem 1. *There exist a separable commutative unital complex Banach algebra $\mathcal{E}$ with dense invertible group, a bounded bijective complex-linear map $\Phi : \mathcal{E} \rightarrow \mathcal{E}$, and a nonzero idempotent $c = d \in \mathcal{E}$ such that*

$$xy = c \implies \Phi(x)\Phi(y) = d,$$

while Φ does not preserve invertibility. Consequently Problem 3.5 of [1] has a negative answer.

Proof. Take the coordinatewise direct product

$$\mathcal{E} = \mathbb{C} \times \mathcal{A}, \quad \|(\alpha, a)\| = |\alpha| + \|a\|.$$

By Lemma 2, $\mathcal{E}$ is separable and commutative, and its invertibles

$$\mathcal{E}^{-1} = \mathbb{C}^\times \times \mathcal{A}^{-1}$$

are dense. Define

$$\Phi(\alpha, a) = (\alpha, L(a)), \quad c = d = (1, 0_{\mathcal{A}}).$$

The map Φ is a bounded linear involution, hence is continuous and bijective. Also $c = d$ is a nonzero idempotent.

Suppose

$$(\alpha, a)(\beta, b) = c.$$

Coordinatewise multiplication gives $\alpha\beta = 1$ and $ab = 0$. Since $\mathcal{A}$ is an integral domain, $a = 0$ or $b = 0$. In either case $L(a)L(b) = 0$, and therefore

$$\Phi(\alpha, a)\Phi(\beta, b) = (\alpha\beta, L(a)L(b)) = (1, 0_{\mathcal{A}}) = d.$$

Thus the fixed-product hypothesis holds for every factorization of c .

Finally, the identity of $\mathcal{E}$ is $u = (1, 1_{\mathcal{A}})$, whereas

$$\Phi(u) = (1, z).$$

The first element is invertible, but the second is not because z is not invertible in $\mathcal{A}$. Hence Φ does not preserve invertibility even though $d \neq 0$. $\square$

5 Verification and scope

The proof is exact and does not use a finite computation. The points most worth checking are: (i) the spectral-radius passage proving radicality; (ii) that radicality makes precisely the nonzero-scalar elements of the unitization invertible; and (iii) that domainhood converts the condition $ab = 0$ into the disjunction $a = 0$ or $b = 0$. These are recorded in detail in the accompanying `verification.md`.

This counterexample answers only Problem 3.5. It does not address Problem 3.4 concerning $\mathcal{B}(X)$.

Bounded novelty check. On 11 August 2026, the local run indexes were searched by arXiv id, title, and the core terms. External exact/close searches combined the wording of Problem 3.5 with “fixed product preserving”, “counterexample”, “integral domain”, “radical Banach algebra”, and “dense invertibles”. The audit included the source and publisher pages and search-indexed portions of Julius’s 2026 survey chapter on fixed-product preservers. No prior explicit answer or matching construction was found. The publisher lists seven citing works, not all of whose full texts were individually inspected. Novelty confidence is therefore moderate; a specialist citation review is recommended. This limitation does not affect the internal validity of the counterexample.

References

- [1] H. Julius, *Fixed product preserving mappings on Banach algebras*, J. Math. Anal. Appl. **517** (2023), 126615; [arXiv:2201.12912](#).
- [2] H. G. Dales, *Banach Algebras and Automatic Continuity*, London Mathematical Society Monographs, New Series 24, Clarendon Press, Oxford, 2000.